

UE24CS352A- Machine Learning

Mini-Project Assignment

CORPORATE BANKRUPTCY PREDICTION USING MACHINE LEARNING

Team Members

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1. Problem Statement

Corporate bankruptcy prediction is a binary classification problem that aims to identify whether a company is likely to become bankrupt using its financial indicators. Early identification of financially distressed companies can help support informed financial decision-making. The objective of this project is to **implement and compare multiple machine learning classification models** for predicting corporate bankruptcy and analyze their performance on a highly imbalanced dataset.

The models implemented are:

- Logistic Regression
- Support Vector Machine (SVM)
- Neural Network (MLP)
- Bernoulli Naive Bayes
- XGBoost
- LightGBM
- Random Forest

A Balanced Random Forest experiment was additionally conducted to study the effect of class imbalance.

2. Dataset Details

The **Polish Companies Bankruptcy Dataset** from the UCI Machine Learning Repository was used. The **third-year forecasting dataset (3year.arff)** was selected.

Dataset Property	Value
Total instances	10,503
Financial features	64
Non-bankrupt companies	10,008
Bankrupt companies	495
Target	Binary classification

The dataset is highly imbalanced, with substantially fewer bankrupt companies.

Data Preprocessing

- The ARFF dataset was loaded and converted into a Pandas DataFrame.
- The target variable was separated from the 64 financial features.
- An **80:20 stratified train-test split** was performed.
- Training set: **8,402 instances**
- Test set: **2,101 instances**
- Missing values were handled using **mean imputation**.
- The imputer was fitted on training data and applied to test data.
- Feature standardization was not applied.

3. Approach and Methodology

The project followed the workflow:

Data Loading → Preprocessing → Train-Test Split → Model Training → Evaluation → Cross-Validation → Imbalance Analysis

The models were evaluated using:

- **Accuracy:** Overall proportion of correct predictions.
- **Log Loss:** Quality of predicted probabilities.
- **Precision:** Correctness of predicted bankrupt cases.
- **Recall:** Ability to detect actual bankrupt companies.
- **F1-score:** Balance between precision and recall.
- **Confusion Matrix:** Class-wise prediction performance.

A **5-fold Stratified Cross-Validation** was performed to evaluate model consistency. Mean imputation was included within the cross-validation pipeline to prevent data leakage.

Because the dataset is highly imbalanced, a **Balanced Random Forest** using class weighting was also evaluated.

4. Implementation Overview

The implementation was developed in **Python** using Pandas, NumPy, SciPy, Scikit-learn, XGBoost and LightGBM.

For each model:

1. The model was trained using the training data.
2. Class predictions and probability estimates were generated.
3. Accuracy and log loss were calculated.

4. Precision, recall and F1-score were calculated for the bankruptcy class.
5. Confusion matrices and training time were recorded.

The same train-test split was used for all models to maintain a consistent comparison.

5. Results and Analysis

Model	Test Acc.	Test Log Loss	CV Acc.	CV Log Loss
Logistic Regression	95.10%	0.222	95.12%	0.219
SVM	95.29%	0.190	95.29%	0.190
Neural Network	95.29%	0.189	95.29%	0.188
Bernoulli Naive Bayes	76.39%	2.326	76.44%	2.442
XGBoost	95.29%	0.178	95.29%	0.175
LightGBM	95.29%	0.178	95.29%	0.174
Random Forest	95.86%	0.190	96.03%	0.171

Balanced Random Forest

The Balanced Random Forest achieved **95.95% accuracy, 89% bankrupt precision, 16% bankrupt recall, and 27% bankrupt F1-score**. Although precision increased compared with the original Random Forest, recall decreased from 18% to 16%. Thus, class weighting alone did not improve minority-class detection in this experiment.

6. Conclusion

This project implemented and compared seven machine learning models for corporate bankruptcy prediction using financial indicators.

The experiments showed that:

- Random Forest produced the highest observed overall and cross-validation accuracy.
- The strong class imbalance significantly affected the detection of bankrupt companies.
- High overall accuracy did not necessarily result in high bankrupt-class recall.
- The Balanced Random Forest experiment showed that class weighting alone was insufficient to improve bankrupt-class recall.
- Therefore, **accuracy should be considered together with precision, recall and F1-score**, especially when evaluating the minority bankruptcy class.